

COOCH BEHAR GOVERNMENT ENGINEERING COLLEGE

Department of Computer Science & Engineering

ASSIGNMENT-1 [CSE 3RD SEMESTER]

IT WORKSHOP LAB [Python Programming]

1. Write a program that swaps the values of variables M and N. You are not allowed to use a third variable. You are not allowed to perform arithmetic on M and N.
2. Assume a suitable value for temperature of a city in Fahrenheit degrees. Write a program to convert this temperature into centigrade degrees and print both temperature.
3. Any year is input through the keyboard. Write a program to determine whether the year is a leap year or not.
4. Write a program to find out the minimum of three numbers.
5. Given 3 points (x1, y1), (x2, y2) and (x3, y3) Write a program to check if all the 3 points fall on straight line.
6. Given the coordinates (x, y) of center of a circle and its radius, write a program that will determine whether a point lies inside the circle, on the circle or outside the circle.
7. Write a program to calculate the distance between two points (x₁,y₁) & (x₂,y₂).
8. Write a program to calculate grade of the MAKAUT result.
9. Write a program to calculate the menu driven calculator that performs:
 1. ADD 2. SUB 3. MULT 4. DIV
10. Write a program to find the sum of the digits of a number.
11. Write a program to find the reverse of a given number
12. Write a program to find the Fibonacci series.
13. Write a program to display following patterns.

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*  
* *  
* * *  
* * * *
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14. Write a program to print all prime numbers from 1 to 300.
15. Write a program to print all Armstrong numbers between 1 and 500.
16. Write a program to generate all Pythagorean Triplets with side length less than or equal to 30.
17. Write a program to implement linear search.
18. Write a program to implement factorial of a number using recursion.
19. Write a program to implement a^b using recursion.
20. Write a program to implement sum of list of integers.
21. Write a program to implement a Stack data structure.
22. Write a program implement a Queue data structure.
23. Suppose a list has 15 numbers. Write a program that removes all duplicates from this list.
24. Suppose a date is represented as a tuple (d,m,y). Write a program to create two date tuples and find the number of days between the two dates.

25. Write a program to carry out the following operations on the given set

$S = \{12, -2, 56, 1, 67, 3\}$

- number of items in set S
- maximum element in set S
- minimum element in set S
- sum of all element in set S
- obtain a new sorted set from S, set S remaining unchanged
- report whether 40 is a member of the set
- report whether 1 is an element of set S

26. Create a list of cricketers. Use this list to create a dictionary in which the list value become key values of the dictionary. Set the values of all keys to 50 in the dictionary created.

27. Write a program to implement union, intersection, set difference, symmetric difference of two given sets

28. Write a program to sort a dictionary in ascending / descending order by value.